

JOHN K. RYU

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PROFESSIONAL EXPERIENCE

AI Engineer | TecAce Software Ltd

Bellevue, WA | April 2025 - Present

- Architected enterprise-grade AI chatbot builder platform with a retrieval-augmented generation (RAG) pipeline utilizing LangChain, LangSmith, and LangGraph for advanced document processing and retrieval
- Engineered LangGraph-based agentic workflow platform — visual flow builder with intelligent routing, guardrails, conversation memory, and node-level inference overrides powering multi-agent enterprise applications
- Built messaging-platform integration gateway connecting Slack and Microsoft Teams to AI applications with signed webhooks, multi-app alias routing, and real-time SSE streaming replies
- Developed hybrid search optimization implementing RRF reranking, cross-encoder reranking, and score fusion techniques for enhanced search relevance across Milvus vector databases
- Delivered external developer platform — access-key-secured public API gateway plus a complete data-driven API reference documentation portal covering 500+ endpoints across six microservices
- Championed AI-native engineering practice orchestrating coding agents (Claude Code) through spec → plan → multi-agent implementation → adversarial QA workflows across a 12-repository platform

e-Commerce Engineer/Owner | Roem Ventures LLC

Seattle, WA | May 2022 - Present

- Founded and developed e-commerce brand from concept to market launch including comprehensive product development, logistics coordination, and business operations management
- Engineered custom Shopify solutions using Liquid programming developing tailored plugins and functionality enhancements for optimized user experience
- Implemented integrated marketing campaign strategies utilizing Google AdWords and Facebook Advertising platforms for targeted customer acquisition and conversion optimization

Software Engineer - Xfinity X1 & Flex | Comcast

Philadelphia, PA | Oct 2017 - Nov 2021

- Successfully developed and maintained Xfinity Flex Streaming product from ground up building key middleware APIs that power core Flex features, resulting in positive customer feedback
- Exceeded subscription acquisition targets by 1000% surpassing initial goal of 100K subscriptions within six months by acquiring over 1 million subscriptions
- Developed and maintained core features for cloud-based X1 Guide interface directly impacting over 31 million customers across Xfinity TV platform
- Implemented scalable backend architecture using distributed systems design principles to support millions of concurrent streaming users

Startup Founder | Parkgorithm

Philadelphia, PA | May 2016 - Jun 2017

- Founded startup company with vision to resolve urban street parking challenges through innovative mobile technology solutions
- Led cross-functional development team managing product roadmap and technical implementation for Android mobile application MVP

Software Engineer | Graphite GTC

Bryn Mawr, PA | May 2015 - May 2016

- Developed client-side applications for Android and Web platforms implementing responsive user interfaces and cross-platform compatibility solutions
- Contributed to development of proprietary Graphite software platform focusing on user experience optimization and performance enhancement

Undergraduate Researcher | Temple University

Philadelphia, PA | Jun 2015 - Oct 2015

- Conducted computer vision research under NSF funding developing machine learning algorithms for mobile robot object detection and recognition systems
- Developed innovative "Supported_By" algorithm in MATLAB enabling spatial relationship detection for autonomous robotics applications

Founder/Sales Director | Fatty Pocket

Norristown, PA | Sept 2011 - Jul 2013

- Founded and scaled digital marketing platform collaborating with local businesses to develop and execute targeted coupon marketing campaigns across multiple industries
- Led business development and sales operations managing marketing professionals and sales teams to identify growth opportunities and maintain strategic vendor relationships

TECHNICAL SKILLS

Agentic / AI-Native Development:

Claude Code • Coding Agents (Codex) • Multi-Agent Orchestration • MCP (Model Context Protocol) • Prompt Engineering • Agent-Assisted SDLC • LangGraph Workflows

Frontend:

React.js • Next.js 14 • TypeScript • JavaScript ES6+ • HTML5/CSS3 • Tailwind CSS • Responsive Design

Databases:

PostgreSQL • MySQL • DynamoDB • MongoDB • Vector Databases (Milvus) • Redis/Valkey

AI/Machine Learning:

RAG Pipeline Architecture • Hybrid Search Systems • Vector Databases • LangChain • PyTorch • AI Chatbot Development • Intent Classification • Cross-encoder Reranking • Semantic Search • SSE Streaming • vLLM/On-Prem Model Serving

Backend:

Python • FastAPI/Flask • Java • Spring Boot • Node.js/Express • RESTful APIs • GraphQL • Microservices • OpenAPI/Swagger

Cloud/DevOps:

AWS (EC2, S3, Lambda, EKS, Bedrock) • Docker • Kubernetes • CI/CD Pipelines • GitHub Actions • Traefik • Observability/Tracing

CERTIFICATIONS

Anthropic — Claude Certified Architect: Foundational (CCA-F) | Issued June 5, 2026 · Valid through June 5, 2027 · Verify: credly.com/users/john-ryu

KEY PROJECTS

AI Chat Builder Platform - Enterprise Agentic RAG System

Hybrid search (BM25 sparse + dense vectors), LangGraph agent workflows, multi-platform chat integrations. Tech: Python, FastAPI, LangChain/LangGraph, AWS Bedrock, Milvus

MarketScopeAI (omnidora.com) - AI Marketing Automation SaaS

AI employees that qualify leads, nurture prospects, and run marketing playbooks 24/7 for small businesses. Tech: Next.js, TypeScript, AI Agents

PlinthPrep (plinthprep.com) - CCA-F Exam Prep Platform

Practice-exam platform for Anthropic's CCA-F certification with AI-generated question variations, cheat sheets, and decision trees. Tech: Next.js, Claude API

RecompIQ (recompiq.com) - Health Data Coaching App

Body-recomposition tracker for protocols, nutrition, biomarkers, and workouts with evidence-graded insights and projections. Tech: Next.js, TypeScript, Data Visualization

EDUCATION

Bachelor of Science in Computer Science

Temple University | College of Science & Technology | Philadelphia, PA